

KOPI

INTERNAL REPORT

USER MANUAL FOR THE
COMPUTER PROGRAM CAPT

52406-49 1st December, 1985

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COMPUTER PROGRAM CAPT

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1. INTRODUCTION

CAPT is a special version of CAP for calculation of vertical bearing capacity of rectangular foundations on soil with (linearly) increasing shear strength with depth. When applied to other soil profiles one should always check the result against other bearing capacity solutions from the "CAP Family" or theoretical bearing capacity formulas.

2. VERTICAL BEARING CAPACITY FOR LINEARLY INCREASING SHEAR STRENGTH WITH DEPTH

a) Surface, strip load, theoretical solution

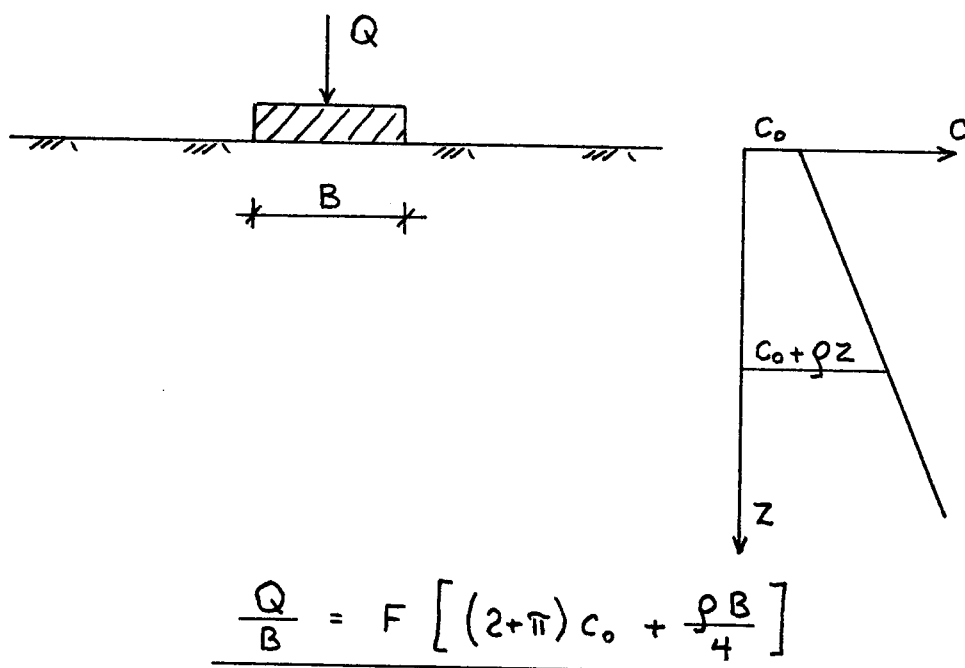


Fig. 1. Bearing capacity, Davis & Booker.

Davis and Booker (Ref. 1) claim that the above solution, with the factor F taken from the diagram below, gives an exact solution for the bearing capacity of a strip load foundation placed on the surface of a soil with linearly increasing shear strength with depth.

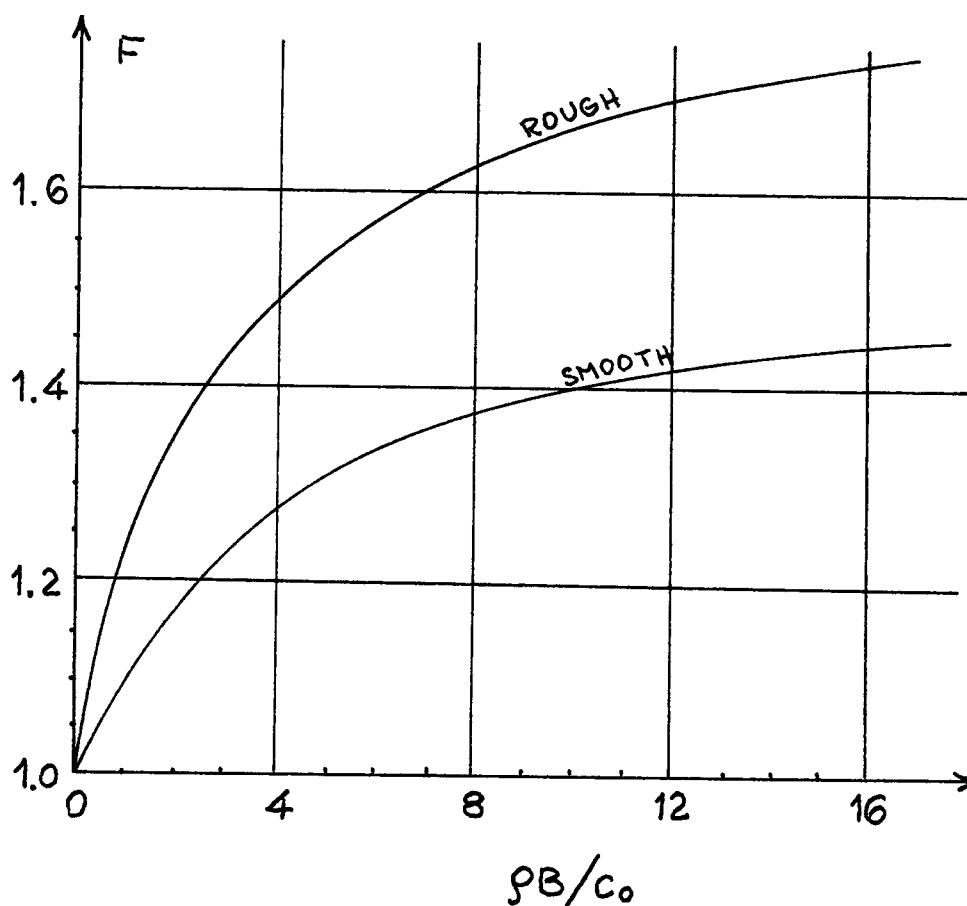


Fig. 2. Correction factor for Davis & Booker's Formula.

The method Davis & Booker has applied in their solution is "LINES OF CHARACTERISTICS", and examples of stress fields achieved with their method are given in Fig. 3 and 4.

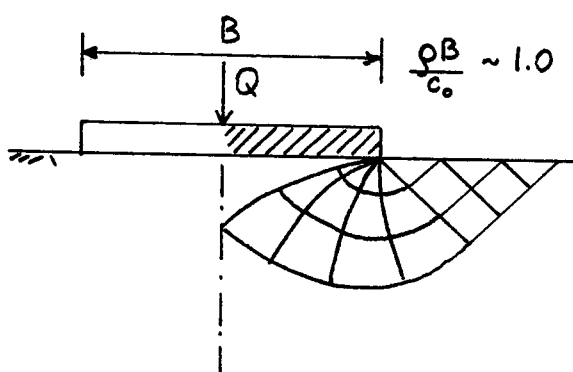


Fig. 3 Stress field for narrow rough footing. Davis & Booker.

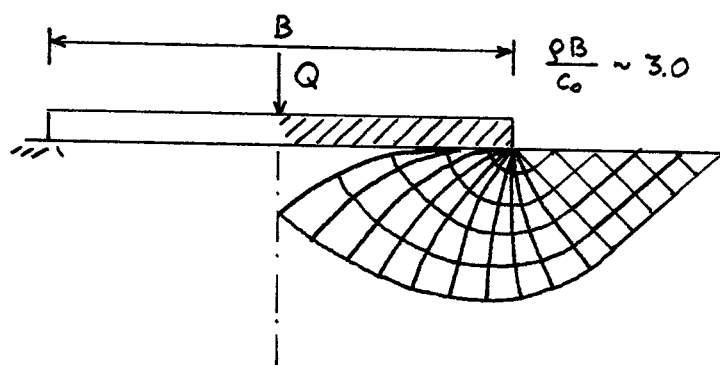


Fig. 4 Stress field for wide rough footing. Davis & Booker.

b) General solution

Davis & Booker's solution cannot take into account soil anisotropy, effect of skirts or effect of foundation shape. A variation of the limit equilibrium program CAP, called CAPT, was made for this purpose.

Only rough footing is of interest, because most offshore gravity platforms are equipped with skirts. CAPT is calibrated to fit the Davis & Booker - rough solution for strip surface loading.

The failure mode analysed in CAPT is shown in Figure 5. Each half

failure figure is a CAP slip surface starting at the platform centre at skirt tip level, with an additional horizontal friction force between platform and soil at skirt tip level.

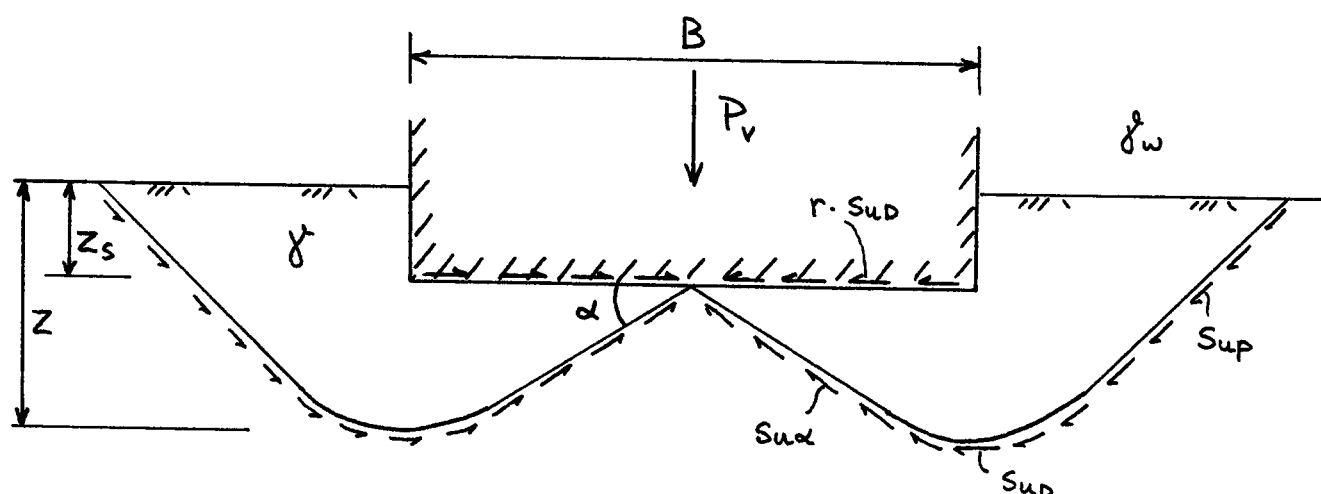


Fig. 5 Failure mode with CAPT

Mobilized horizontal shear stress, $r \cdot s_{uD}$, at skirt tip level, with a roughness factor $r = 0.35$ gives very good agreement between CAPT and Davis and Booker's theoretical solution for surface strip loads. In fact, CAPT results in F-values of the order of 1 percent on the safe side as compared to Davis & Booker's solution.

3. CAPT AS GENERAL PROGRAM FOR VERTICAL BEARING CAPACITY CALCULATIONS

The method developed under 2 b) has been used for anisotropic, inhomogeneous, layered soil for geometries with rectangular area and long skirts - and the results seems to be reasonable.

The input data form for the CAP program, see 52406-46 or 52406-47, is used, with the following modifications:

THIRD LINE (PLATFORM BASE GEOMETRY)

- | | | |
|-------|-----|---|
| 01-10 | B | - Always use half the platform width, $B/2$, instead of B. |
| 21-30 | ZSS | - If $ZS = 0.0$ the program sets $ZS = 0.001$. |
| 71-80 | PS | - Should be left blank. |

FOURTH LINE (FORCES ON PLATFORM)

- | | | |
|-------|-------|--|
| 01-10 | PVS | - Recommended value = 500, which represents $PV/2$. |
| 11-20 | PVF | " " " " " " |
| 31-40 | PH | - Use 0.0 |
| 41-50 | M | - Use 0.0 |
| 51-60 | DELPH | - Use 0.0, i.e. no wave activity |
| 61-70 | DELPT | " " " " " " |

71-80 FAC - Use 1.0

FIFTH LINE (SLIP SURFACE PARAMETERS)

11-20	ZSTEP	- The first depth is (ZS + ZSTEP), and not ZS as in CAP.
21-30	REDBS	- Use 0.0
31-40	REDSS	- Recommended value = 0.5, which represents a positive shape factor. If there is doubt about this effect, use REDSS = 0.0
41-50	CRACK	Use 0.0
51-60	PASTYP	Use 1.0

All other input-data is covered in the regular CAP-manual.

The result of the computer calculation is a safety factor, SF. Since there is only one force, PV, these two figures must be proportional to each other. The vertical bearing capacity of the structure is therefore: $QV = SF * PV * 2$, or if the recommended value is used: $QV = 1000 SF$.

PROGRAM "FLOW CHART" FOR CAPT

MAIN

INPUT2

CALC2

ARRAY

RPRINT

SUBS

FACT

SHEAR2

SUBS

SAF2

SHEAR2

SUBS

FORCE

FACT

FORCC2

HHEAD

SHEAR2

SUBS

FORCE

FACT

HEAD

HANSEN

MEYER

FORCE2

HHEAD

SHEAR2

SUBS

FORCE

FACT

HEAD

OUTPUT

OUTPUT

CHANGES WITH RESPECT TO CAP

A lot of editorial changes has been made because of the change of program name and the fact that some options or parameters are not needed in the CAPT program. These changes are of minor interest to the user of the program, but can be found by comparing the listings of the two programs.

The main differences are in subroutine FORCE.

CAP

$$BO = BB - 2. * E$$

$$A(3) = -2. * E * AL$$

$$B(5) = -(DELPT + 0.5 * GAM1 * ZS) * ZS * AL$$

CAPT

$$BO = BB$$

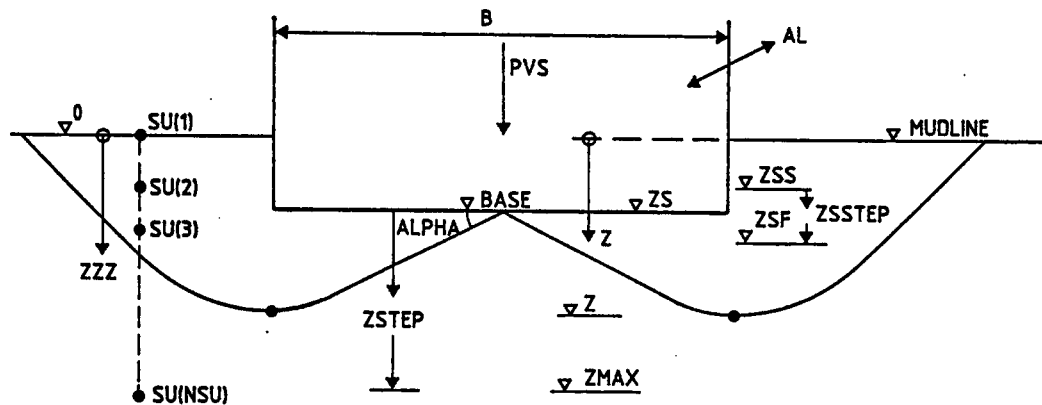
$$A(3) = -0.35 * BB * AL$$

$$B(5) = -DELPT * ZS * AL$$

Reference is made to alphabetic list of symbols used in CAP, see report 52406-47.

BEARING CAPACITY CALCULATED BY SLIP SURFACES

INPUT
DATA FORM
FOR CAPT



1	57
PROJECT	

1	30
UNITS USED	

[illegible]

10 11		20 21		30 31		40 41		50 51		60 61		70 71		80	
0.5* PVS		0.5* PVF		PVSTEP		PH		AM		DELPH		DELPT		FAC 2)	
					0.0		0.0		0.0		0.0		0.0		1.0

10 11		20 21		30 31		40 41		50 51		60 61		70 71		80	
ZMAX	ZSTEP	REDBS 3)	REDSS 4)	CRACK 9)	PASTYP 10)	ZSPEC 1 4)	ZSPEC 2 4)								
		0.0			0.0	1.0									

[illegible]

	1	10 11	20 21	30 31	40 41	50
	ZZZ	SU ⁵⁾	SUD ⁶⁾	SUP ⁷⁾	SU2 ¹¹⁾	
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						

- 1) TOTAL UNIT WEIGHT OF SOIL AND WATER
- 2) LOAD FACTOR ON PH, M, ΔPH AND $\Delta PT = 1.0$
- 3) REDUCTION FACTOR FOR SIDE SHEAR,
BASE/SOIL=0.0
- 4) REDUCTION FACTOR FOR SIDE SHEAR,
SOIL/SOIL. NORMALLY 0.5
- 5) TRIAXIAL COMPRESSION STRENGTH, OR SU
- 6) SIMPLE SHEAR STRENGTH
- 7) TRIAXIAL EXTENSION STRENGTH
- 8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
NPR1 $\neq$ 0 GIVES TABLE OF SF-VALUES
NPR2 $\neq$ 0 GIVES ALL K (N) FOR SFMIN
NPR3 $\neq$ 0 OMITTED IN THIS MODE
NPR4 $\neq$ 0 OMITTED IN THIS MODE
NPR5 $\neq$ 0 OMITTED IN THIS MODE
- 9) CRACK ON HEEL SIDE: $\neq$ 0. OPEN
- 10) TYPES OF PASSIVE FAILURE: $\neq$ 1.0 BIG AREA
- 11) MINIMUM HORIZONTAL STRENGTH

#) MAY BE LEFT BLANK.

////// INPUT DATA OMITTED IN THIS FAILURE MODE.

LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.

BLANK LINE ENDING ALL DATA SETS.



OPPDRAAG :
JOB TITLE:

OPPDRAAGSNR.
CONTRACT NO.
52406-48

SIDE
PAGE I-0

FORSØK/BEREGNING:
SUBJECT:

The Computer Program CAPT

SIGN. RAL

DATO
DATE 10/12-8

KONTR.
CONTR.

DATO
DATE

APPENDIX I : EXAMPLES



OPPDRAAG :
JOB TITLE:

OPPDRAAGSNR.
CONTRACT NO.
52406-48

SIDE
PAGE
I - 1

FORSØK / BEREGNING:
SUBJECT:

SIGN. *RAL*

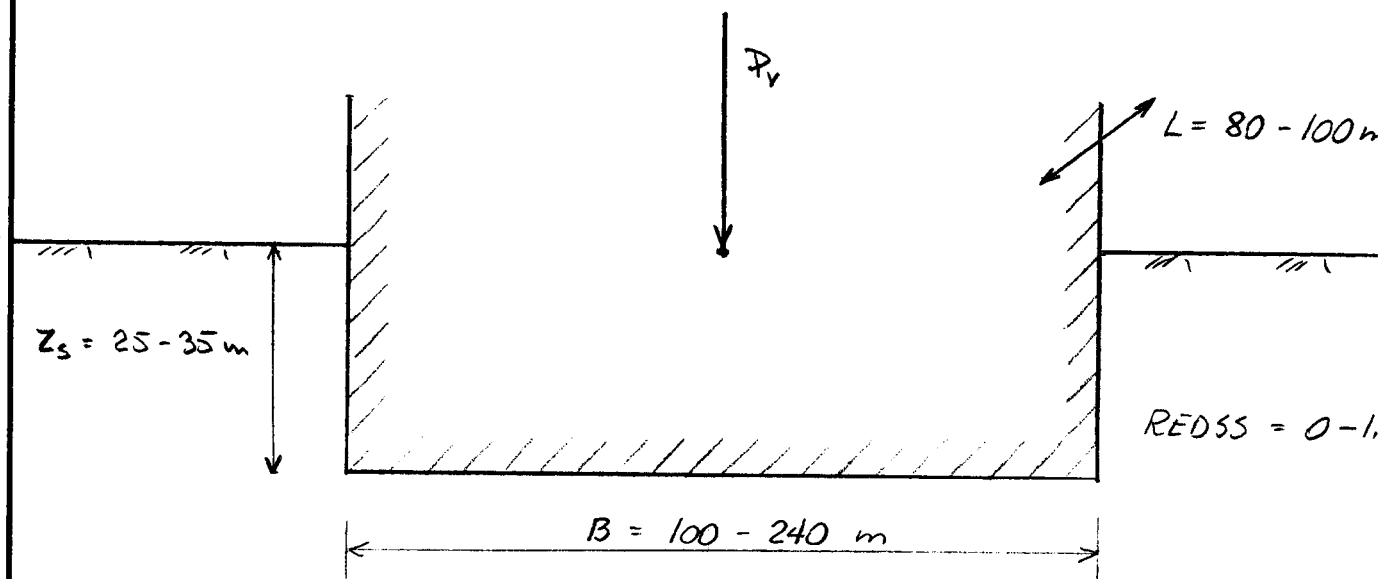
DATO
DATE 10/12-8

DATA FOR CAPT TEST EXAMPLES

KONTR.
CONTR.

DATO
DATE

Summary of data variations:

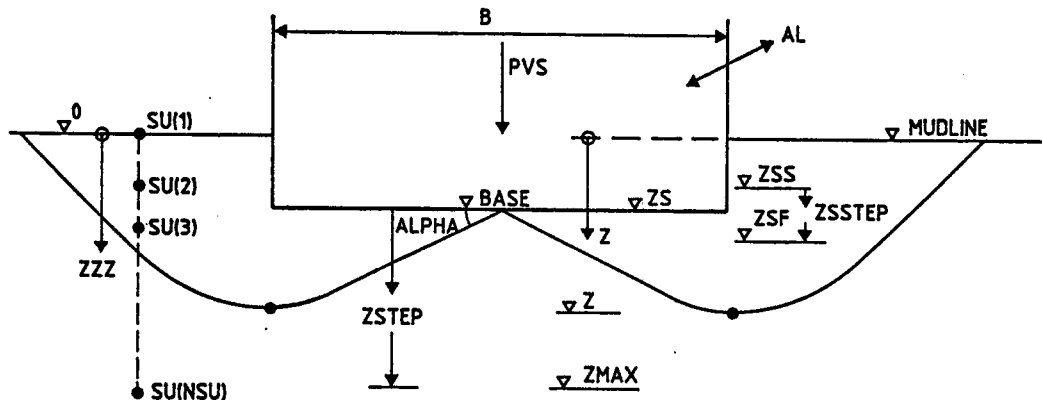




JOB TITLE:	CAP - FAMILY EXAMPLES	CONTRACT NO.	52406-48	PAGE	I-2
SUBJECT:	FUNCTION CONTROL FOR CAPT	SIGN.	RAL / P.T.	DATE	1/12/85
		CONTR.		DATE	

Function to be checked in RUN NO.		1	2	3	4	5	6
1. ZSTEP	, calculation for z_s - values		X	X			
2. GAMS	, calculation independent of δ -value or not?			X			
3. PVSTEP	, calculation for P_v - values			X			
4. REDSS	, reduction factor soil/soil	.5	.5	.5	.0	.5	1.
5. ZSPEC 1 & 2	, specified failure surfaces, K(N) table					X	
6. NPR 1	, 0, 1 Table SF/Z	1	1	0	1	1	0
7. NPR 2	, 0, 1 Table K(N) for SF _{min}	1	0	0	1	1	1
8. BMAX/DELB	, more areas in one run						X
9. SU	, shear strength options		X				
10. SU/SU2	, — " —			X			
11. SU, SUD, SUP	, — " —					X	
12. SU, SUD, SUP, SU2	, — " —						X
13. More datasets	, in one run			X			

INPUT
DATA FORM
FOR CAPT

[illegible]

7) MAY BE LEFT BLANK.
 //// INPUT DATA OMITTED IN THIS FAILURE MODE.
 LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.
 BLANK LINE ENDING ALL DATA SETS.

C A P T

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 1 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION

B = 50.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 30.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSTEP = 0.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.020

WATER UNIT WEIGHT

GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 0.000

HORIZONTAL FORCE AT MUDLINE

PH = 0.000

OVERTURNING MOMENT AT MUDLINE

M = 0.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.000

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = 0.000

LOAD FACTORS TO BE USED

LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.000

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.500

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.200

PROJECT : EXAMPLE 1 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
32.000	2.101	42.001	-0.8400E+03	0.2000E+02
34.000	4.200	22.631	-0.9052E+03	0.4000E+02
36.000	6.302	16.212	-0.9727E+03	0.6000E+02
38.000	8.410	13.050	-0.1044E+04	0.8000E+02
40.000	10.527	11.194	-0.1119E+04	0.1000E+03
42.000	12.657	9.993	-0.1199E+04	0.1200E+03
44.000	14.804	9.207	-0.1289E+04	0.1400E+03
46.000	16.972	8.662	-0.1386E+04	0.1600E+03
48.000	19.164	8.272	-0.1489E+04	0.1800E+03
50.000	21.386	7.992	-0.1598E+04	0.2000E+03
52.000	23.642	7.791	-0.1714E+04	0.2200E+03
54.000	25.938	7.650	-0.1836E+04	0.2400E+03
56.000	28.280	7.556	-0.1965E+04	0.2600E+03
58.000	30.677	7.499	-0.2100E+04	0.2800E+03
60.000	33.132	7.471	-0.2241E+04	0.3000E+03
62.000	35.659	7.467	-0.2389E+04	0.3200E+03
64.000	38.269	7.483	-0.2544E+04	0.3400E+03
66.000	40.977	7.515	-0.2706E+04	0.3600E+03
68.000	43.801	7.562	-0.2874E+04	0.3800E+03

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 7.467
Z = 62.000
ALPHA = 35.659

i.e. Vertical bearing capacity :

$$\underline{P_{VMAX} = 7,467 \text{ MN}}$$

PROJECT : EXAMPLE 1 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 7.467
 FOUND FOR DEPTH Z = 62.000
 AND SLIP SURFACE ANGLE ALPHA = 35.659

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.005	0.0000E+00
3	-RH	-0.1423E+02	-0.1750E+04	0.061	0.008	0.0000E+00
4	HA	0.1679E+04	-0.7953E+04	0.106	0.014	0.1792E+04
5	-PA	-0.2877E+02	-0.7712E+04	0.028	0.004	0.0000E+00
6	-HP	-0.1596E+04	-0.8227E+04	0.113	0.015	-0.1472E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.004	0.0000E+00
8	-RS2	-0.3064E+02	-0.2283E+04	0.100	0.013	0.0000E+00
9	-RS3	-0.9008E+01	-0.2180E+04	0.031	0.004	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.003	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.003	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.003	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.005	0.0000E+00

PROJECT : EXAMPLE 1 ***CAPT***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

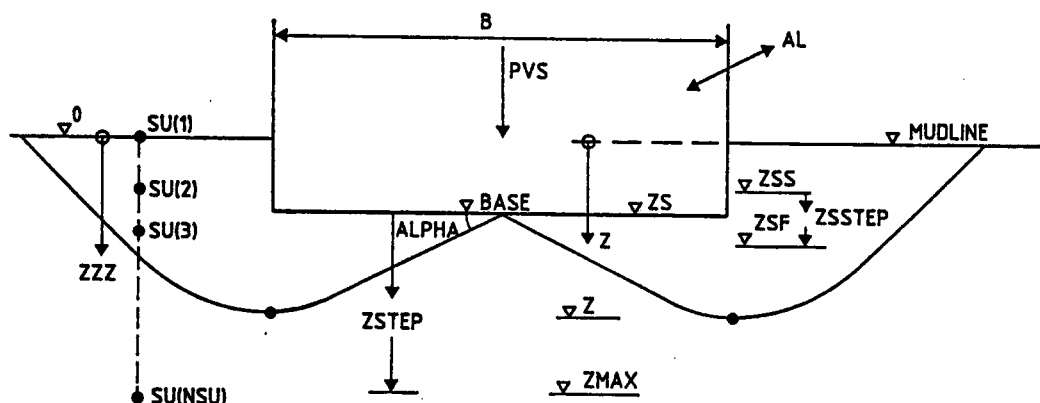
LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	7.47	62.00	35.7	0.0106	0.0085	0.0090

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

BEARING CAPACITY CALCULATED BY SLIP SURFACES

INPUT
DATA FORM
FOR CAPT



PROJECT	57
EXAMPLE 2 *** CAPT ***	

1	30
UNITS USED	
SI - UNITS MN & M	

1										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
0.5* B										AL										ZSS										ZSF										ZSSTEP										GAMS ¹⁾										GAMW ¹⁾										PS ^{*)}																			
50.0										100.0										25.0										35.0										5.0										0.020										0.010																													

1	10 11	20 21	30 31	40 41	50 51	60 61	70 71	80 81
0.5* PVS	0.5* PVF	PVSTEP	PH	AM	DELPH	DELPT	FAC	ZI
500.0	500.0	0.0	0.0	0.0	0.0	0.0	1.0	

10 11		20 21		30 31		40 41		50 51		60 61		70 71		80	
ZMAX		ZSTEP		REDBS ³⁾		REDSS ⁴⁾		CRACK ⁹⁾		PASTYP ¹⁰⁾		ZSPEC 1 ⁸⁾		ZSPEC 2 ⁸⁾	
70.0		2.0		0.0		0.5		0.0				1.0			

[illegible]

1	10 11	20 21	30 31	40 41	50
	ZZZ	SU ⁵⁾	SUD ⁶⁾	SUP ⁷⁾	SU2 ¹¹⁾
1	0.0	0.001			
2	100.0	0.225			
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

- 1) TOTAL UNIT WEIGHT OF SOIL AND WATER
- 2) LOAD FACTOR ON PH, M, ΔPH AND $\Delta PT = 1.0$
- 3) REDUCTION FACTOR FOR SIDE SHEAR,
BASE/SOIL = 0.0
- 4) REDUCTION FACTOR FOR SIDE SHEAR,
SOIL/SOIL, NORMALLY 0.5
- 5) TRIAXIAL COMPRESSION STRENGTH, OR SU
- 6) SIMPLE SHEAR STRENGTH
- 7) TRIAXIAL EXTENSION STRENGTH
- 8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
NPR1 = 0 GIVES TABLE OF SF-VALUES
NPR2 = 0 GIVES ALL K (N) FOR SFMIN
NPR3 = 0 OMITTED IN THIS MODE
NPR4 = 0 OMITTED IN THIS MODE
NPRS = 0 OMITTED IN THIS MODE
- 9) CRACK ON HEEL SIDE: = 0. OPEN
- 10) TYPES OF PASSIVE FAILURE: = 1.0 BIG AREA
- 11) MINIMUM HORIZONTAL STRENGTH

* MAY BE LEFT BLANK.

//// INPUT DATA OMITTED IN THIS FAILURE MODE.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1

C A P T

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 2 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION

B = 50.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 25.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 35.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSTEP = 5.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.020

WATER UNIT WEIGHT

GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 0.000

HORIZONTAL FORCE AT MUDLINE

PH = 0.000

OVERTURNING MOMENT AT MUDLINE

M = 0.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.000

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = 0.000

LOAD FACTORS TO BE USED

LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.000

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.500

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EXAMPLE 2 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 25.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
27.000	2.161	37.101	-0.7420E+03	0.2000E+02
29.000	4.322	20.083	-0.8033E+03	0.4000E+02
31.000	6.486	14.469	-0.8681E+03	0.6000E+02
33.000	8.656	11.715	-0.9372E+03	0.8000E+02
35.000	10.838	10.109	-0.1011E+04	0.1000E+03
37.000	13.034	9.115	-0.1094E+04	0.1200E+03
39.000	15.249	8.454	-0.1184E+04	0.1400E+03
41.000	17.487	7.998	-0.1280E+04	0.1600E+03
43.000	19.753	7.680	-0.1382E+04	0.1800E+03
45.000	22.051	7.458	-0.1492E+04	0.2000E+03
47.000	24.388	7.309	-0.1608E+04	0.2200E+03
49.000	26.769	7.213	-0.1731E+04	0.2400E+03
51.000	29.202	7.159	-0.1861E+04	0.2600E+03
53.000	31.693	7.139	-0.1999E+04	0.2800E+03
55.000	34.254	7.147	-0.2144E+04	0.3000E+03
57.000	36.896	7.176	-0.2296E+04	0.3200E+03
59.000	39.633	7.225	-0.2456E+04	0.3400E+03
61.000	42.484	7.289	-0.2624E+04	0.3600E+03

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 7.139
Z = 53.000
ALPHA = 31.693

PROJECT : EXAMPLE 2 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
32.000	2.101	48.548	-0.9709E+03	0.2000E+02
34.000	4.200	26.097	-0.1044E+04	0.4000E+02
36.000	6.301	18.655	-0.1119E+04	0.6000E+02
38.000	8.408	14.986	-0.1199E+04	0.8000E+02
40.000	10.524	12.831	-0.1283E+04	0.1000E+03
42.000	12.653	11.434	-0.1372E+04	0.1200E+03
44.000	14.799	10.516	-0.1472E+04	0.1400E+03
46.000	16.965	9.878	-0.1581E+04	0.1600E+03
48.000	19.156	9.420	-0.1696E+04	0.1800E+03
50.000	21.377	9.087	-0.1817E+04	0.2000E+03
52.000	23.631	8.848	-0.1947E+04	0.2200E+03
54.000	25.926	8.678	-0.2083E+04	0.2400E+03
56.000	28.266	8.563	-0.2226E+04	0.2600E+03
58.000	30.661	8.492	-0.2378E+04	0.2800E+03
60.000	33.114	8.455	-0.2536E+04	0.3000E+03
62.000	35.639	8.446	-0.2703E+04	0.3200E+03
64.000	38.247	8.462	-0.2877E+04	0.3400E+03
66.000	40.952	8.497	-0.3059E+04	0.3600E+03
68.000	43.775	8.550	-0.3249E+04	0.3800E+03

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 8.446
Z = 62.000
ALPHA = 35.639

PROJECT : EXAMPLE 2 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 35.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
37.000	2.031	61.624	-0.1232E+04	0.2000E+02
39.000	4.061	32.970	-0.1319E+04	0.4000E+02
41.000	6.091	23.429	-0.1406E+04	0.6000E+02
43.000	8.126	18.705	-0.1496E+04	0.8000E+02
45.000	10.169	15.915	-0.1592E+04	0.1000E+03
47.000	12.223	14.096	-0.1692E+04	0.1200E+03
49.000	14.291	12.833	-0.1797E+04	0.1400E+03
51.000	16.377	11.968	-0.1915E+04	0.1600E+03
53.000	18.485	11.348	-0.2043E+04	0.1800E+03
55.000	20.619	10.888	-0.2178E+04	0.2000E+03
57.000	22.783	10.544	-0.2320E+04	0.2200E+03
59.000	24.982	10.289	-0.2469E+04	0.2400E+03
61.000	27.221	10.102	-0.2627E+04	0.2600E+03
63.000	29.507	9.970	-0.2792E+04	0.2800E+03
65.000	31.848	9.881	-0.2964E+04	0.3000E+03
67.000	34.247	9.828	-0.3145E+04	0.3200E+03
69.000	36.716	9.805	-0.3334E+04	0.3400E+03
70.000	37.981	9.803	-0.3431E+04	0.3500E+03

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 9.803
Z = 70.000
ALPHA = 37.981

PROJECT : EXAMPLE 2 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

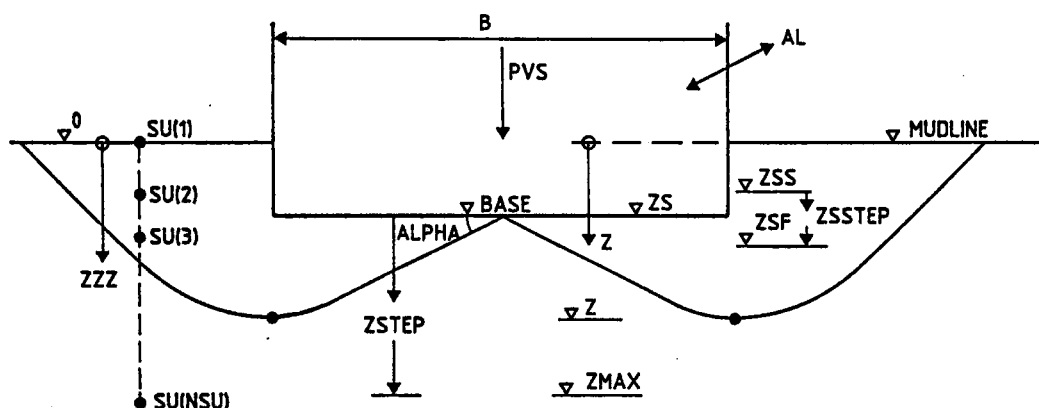
AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	25.0	7.14	53.00	31.7	0.0107	0.0108	0.0113
1.00	500.0	30.0	8.45	62.00	35.6	0.0106	0.0085	0.0090
1.00	500.0	35.0	9.80	70.00	38.0	0.0104	0.0060	0.0063

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

INPUT
DATA FORM
FOR CAPT

[illegible]

////// INPUT DATA OMITTED IN THIS FAILURE MODE.

C A P T

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 4 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION

B = 50.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 30.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSTEP = 0.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.020

WATER UNIT WEIGHT

GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 0.000

HORIZONTAL FORCE AT MUDLINE

PH = 0.000

OVERTURNING MOMENT AT MUDLINE

M = 0.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.000

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = 0.000

LOAD FACTORS TO BE USED

LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.000

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.500

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

1/ 4

PROJECT : EXAMPLE 4 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= $S(A*SA)/(-S(B))$	
			S(A*SA)	S(B)
32.000	2.101	48.548	-0.9709E+03	0.2000E+02
34.000	4.200	26.097	-0.1044E+04	0.4000E+02
36.000	6.301	18.655	-0.1119E+04	0.6000E+02
38.000	8.408	14.986	-0.1199E+04	0.8000E+02
40.000	10.524	12.831	-0.1283E+04	0.1000E+03
42.000	12.653	11.434	-0.1372E+04	0.1200E+03
44.000	14.799	10.516	-0.1472E+04	0.1400E+03
46.000	16.965	9.878	-0.1581E+04	0.1600E+03
48.000	19.156	9.420	-0.1696E+04	0.1800E+03
50.000	21.377	9.087	-0.1817E+04	0.2000E+03
52.000	23.631	8.848	-0.1947E+04	0.2200E+03
54.000	25.926	8.678	-0.2083E+04	0.2400E+03
56.000	28.266	8.563	-0.2226E+04	0.2600E+03
58.000	30.661	8.492	-0.2378E+04	0.2800E+03
60.000	33.114	8.455	-0.2536E+04	0.3000E+03
62.000	35.639	8.446	-0.2703E+04	0.3200E+03
64.000	38.247	8.462	-0.2877E+04	0.3400E+03
66.000	40.952	8.497	-0.3059E+04	0.3600E+03
68.000	43.775	8.550	-0.3249E+04	0.3800E+03

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 8.446
Z = 62.000
ALPHA = 35.639

PROJECT : EXAMPLE 4 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 8.446
 FOUND FOR DEPTH Z = 62.000
 AND SLIP SURFACE ANGLE ALPHA = 35.639

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00
3	-RH	-0.1413E+02	-0.1750E+04	0.068	0.008	0.0000E+00
4	HA	0.1689E+04	-0.7955E+04	0.109	0.013	0.1792E+04
5	-PA	-0.3159E+02	-0.7712E+04	0.035	0.004	0.0000E+00
6	-HP	-0.1604E+04	-0.8227E+04	0.136	0.016	-0.1472E+04
7	-RS1	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00
8	-RS2	-0.3045E+02	-0.2284E+04	0.113	0.013	0.0000E+00
9	-RS3	-0.8932E+01	-0.2180E+04	0.035	0.004	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00

PROJECT : EXAMPLE 4 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	8.45	62.00	35.6	0.0106	0.0085	0.0090

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

C A P T

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 4 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION

B = 50.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 30.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSTEP = 0.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.020

WATER UNIT WEIGHT

GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 0.000

HORIZONTAL FORCE AT MUDLINE

PH = 0.001

OVERTURNING MOMENT AT MUDLINE

M = 0.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.000

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = 0.000

LOAD FACTORS TO BE USED

LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.000

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.150	0.150
100.000	0.150	0.150

1/ 4

PROJECT : EXAMPLE 4 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
32.000	2.098	127.902	-0.2558E+04	0.2000E+02
34.000	4.189	66.051	-0.2642E+04	0.4000E+02
36.000	6.276	45.485	-0.2729E+04	0.6000E+02
38.000	8.365	35.240	-0.2819E+04	0.8000E+02
40.000	10.457	29.123	-0.2912E+04	0.1000E+03
42.000	12.558	25.068	-0.3008E+04	0.1200E+03
44.000	14.670	22.192	-0.3107E+04	0.1400E+03
46.000	16.797	20.051	-0.3208E+04	0.1600E+03
48.000	18.945	18.401	-0.3312E+04	0.1800E+03
50.000	21.116	17.092	-0.3418E+04	0.2000E+03
52.000	23.316	16.031	-0.3527E+04	0.2200E+03
54.000	25.550	15.156	-0.3637E+04	0.2400E+03
56.000	27.824	14.422	-0.3750E+04	0.2600E+03
58.000	30.144	13.800	-0.3864E+04	0.2800E+03
60.000	32.517	13.265	-0.3980E+04	0.3000E+03
62.000	34.953	12.801	-0.4096E+04	0.3200E+03
64.000	37.462	12.396	-0.4215E+04	0.3400E+03
66.000	40.055	12.038	-0.4334E+04	0.3600E+03
68.000	42.750	11.720	-0.4454E+04	0.3800E+03

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 11.720
Z = 68.000
ALPHA = 42.750

PROJECT : EXAMPLE 4 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 11.720
 FOUND FOR DEPTH Z = 68.000
 AND SLIP SURFACE ANGLE ALPHA = 42.750

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.1000E-02	0.0000E+00	0.000	0.000	0.1000E-02
2	PA	0.0000E+00	0.0000E+00	0.150	0.013	0.0000E+00
3	-RH	-0.2240E+02	-0.1750E+04	0.150	0.013	0.0000E+00
4	HA	0.2129E+04	-0.8857E+04	0.150	0.013	0.2242E+04
5	-PA	-0.9870E+02	-0.7712E+04	0.150	0.013	0.0000E+00
6	-HP	-0.2008E+04	-0.9769E+04	0.175	0.015	-0.1862E+04
7	-RS1	0.0000E+00	0.0000E+00	0.150	0.013	0.0000E+00
8	-RS2	0.0000E+00	0.0000E+00	0.172	0.015	0.0000E+00
9	-RS3	0.0000E+00	0.0000E+00	0.150	0.013	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.150	0.013	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.150	0.013	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.150	0.013	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.150	0.013	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.150	0.013	0.0000E+00

PROJECT : EXAMPLE 4 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	11.72	68.00	42.8	0.0135	0.0085	0.0090

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

4

6

8

10

C A P T

12

14

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BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

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METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

20

PROJECT : EXAMPLE 4 ***CAPT***

22

INPUT DATA REPRINT

24

UNITS USED : SI-UNITS MN & M

26

WIDTH OF FOUNDATION

B = 50.000

LENGTH OF FOUNDATION

L = 100.000

28

DEPTH OF EMBEDMENT , START VALUE

ZSS = 30.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

30

DEPTH OF EMBEDMENT , STEP VALUE

ZSSSTEP = 0.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.020

32

WATER UNIT WEIGHT

GAMW = 0.010

34

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 500.000

36

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 0.000

HORIZONTAL FORCE AT MUDLINE

PH = 0.000

38

OVERTURNING MOMENT AT MUDLINE

M = 0.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.000

40

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = 0.000

LOAD FACTORS TO BE USED

LF = 1.0

42

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

44

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.000

46

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.000

48

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

50

SHEAR STRENGTH USED :

52

DEPTH BELOW

UNDRAINED

MINIMUM

54

MUDLINE

SHEAR STRENGTH

SHEAR STRENGTH

56

0.000

0.001

0.001

100.000

0.225

0.225

58

1 / 4

60

62

64

66

68

70

BEARING CAPACITY CALCULATED BY SLIP SURFACES

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PROJECT : EXAMPLE 4 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE	BASE/SLIP SURFACE ANGLE	SAFETY FACTOR	SF= S(A*SA) / (-S(B))	
Z	ALPHA	SF	S(A*SA)	S(B)
32.000	2.101	45.886	-0.9177E+03	0.2000E+02
34.000	4.200	24.459	-0.9784E+03	0.4000E+02
36.000	6.302	17.393	-0.1044E+04	0.6000E+02
38.000	8.411	13.919	-0.1114E+04	0.8000E+02
40.000	10.528	11.884	-0.1188E+04	0.1000E+03
42.000	12.659	10.569	-0.1268E+04	0.1200E+03
44.000	14.807	9.667	-0.1353E+04	0.1400E+03
46.000	16.976	9.024	-0.1444E+04	0.1600E+03
48.000	19.169	8.554	-0.1540E+04	0.1800E+03
50.000	21.392	8.206	-0.1641E+04	0.2000E+03
52.000	23.650	7.947	-0.1748E+04	0.2200E+03
54.000	25.948	7.755	-0.1861E+04	0.2400E+03
56.000	28.291	7.614	-0.1980E+04	0.2600E+03
58.000	30.689	7.514	-0.2104E+04	0.2800E+03
60.000	33.150	7.447	-0.2234E+04	0.3000E+03
62.000	35.679	7.406	-0.2370E+04	0.3200E+03
64.000	38.293	7.387	-0.2511E+04	0.3400E+03
66.000	41.004	7.385	-0.2659E+04	0.3600E+03
68.000	43.834	7.400	-0.2812E+04	0.3800E+03

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 7.385
Z = 66.000
ALPHA = 41.004

PROJECT : EXAMPLE 4 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 7.385
 FOUND FOR DEPTH Z = 66.000
 AND SLIP SURFACE ANGLE ALPHA = 41.004

$$K(N) = A(N) * SA(N) / SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.005	0.0000E+00
3	-RH	-0.1616E+02	-0.1750E+04	0.068	0.009	0.0000E+00
4	HA	0.1955E+04	-0.8503E+04	0.115	0.016	0.2088E+04
5	-PA	-0.3613E+02	-0.7712E+04	0.035	0.005	0.0000E+00
6	-HP	-0.1903E+04	-0.9255E+04	0.140	0.019	-0.1728E+04
7	-RS1	0.0000E+00	0.0000E+00	0.035	0.005	0.0000E+00
8	-RS2	0.0000E+00	0.0000E+00	0.116	0.016	0.0000E+00
9	-RS3	0.0000E+00	0.0000E+00	0.035	0.005	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.005	0.0000E+00

PROJECT : EXAMPLE 4 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

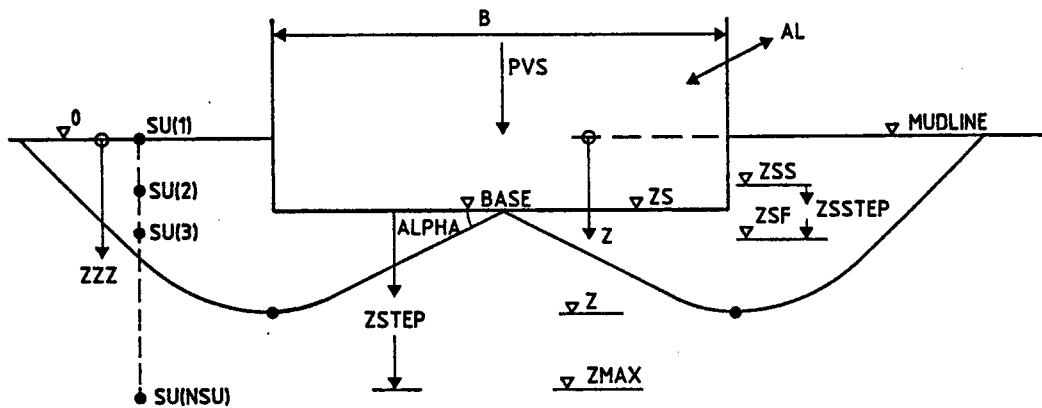
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	7.39	66.00	41.0	0.0132	0.0085	0.0090

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

INPUT
DATA FORM
FOR CAPT

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BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 5 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	50.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.500

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.200

PROJECT : EXAMPLE 5 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
32.000	2.101	42.001	-0.8400E+03	0.2000E+02
34.000	4.200	22.631	-0.9052E+03	0.4000E+02
36.000	6.302	16.212	-0.9727E+03	0.6000E+02
38.000	8.410	13.050	-0.1044E+04	0.8000E+02
40.000	10.527	11.194	-0.1119E+04	0.1000E+03
42.000	12.657	9.993	-0.1199E+04	0.1200E+03
44.000	14.804	9.207	-0.1289E+04	0.1400E+03
46.000	16.972	8.662	-0.1386E+04	0.1600E+03
48.000	19.164	8.272	-0.1489E+04	0.1800E+03
50.000	21.386	7.992	-0.1598E+04	0.2000E+03
52.000	23.642	7.791	-0.1714E+04	0.2200E+03
54.000	25.938	7.650	-0.1836E+04	0.2400E+03
56.000	28.280	7.556	-0.1965E+04	0.2600E+03
58.000	30.677	7.499	-0.2100E+04	0.2800E+03
60.000	33.132	7.471	-0.2241E+04	0.3000E+03
62.000	35.659	7.467	-0.2389E+04	0.3200E+03
64.000	38.269	7.483	-0.2544E+04	0.3400E+03
66.000	40.977	7.515	-0.2706E+04	0.3600E+03
68.000	43.801	7.562	-0.2874E+04	0.3800E+03

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 7.467
Z = 62.000
ALPHA = 35.659

PROJECT : EXAMPLE 5 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 7.467
 FOUND FOR DEPTH Z = 62.000
 AND SLIP SURFACE ANGLE ALPHA = 35.659

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.005	0.0000E+00
3	-RH	-0.1423E+02	-0.1750E+04	0.061	0.008	0.0000E+00
4	HA	0.1679E+04	-0.7953E+04	0.106	0.014	0.1792E+04
5	-PA	-0.2877E+02	-0.7712E+04	0.028	0.004	0.0000E+00
6	-HP	-0.1596E+04	-0.8227E+04	0.113	0.015	-0.1472E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.004	0.0000E+00
8	-RS2	-0.3064E+02	-0.2283E+04	0.100	0.013	0.0000E+00
9	-RS3	-0.9008E+01	-0.2180E+04	0.031	0.004	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.003	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.003	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.003	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.005	0.0000E+00

PROJECT : EXAMPLE 3 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	25.0	6.85	51.00	29.5	0.0104	0.0121	0.0127
1.00	500.0	30.0	8.13	60.00	33.5	0.0104	0.0106	0.0112
1.00	1000.0	25.0	3.42	51.00	29.8	0.0208	0.0256	0.0269
1.00	1000.0	30.0	4.06	60.00	34.1	0.0207	0.0234	0.0245
1.00	1500.0	25.0	2.28	51.00	30.0	0.0313	0.0391	0.0410
1.00	1500.0	30.0	2.70	60.00	34.4	0.0312	0.0361	0.0379

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

C A P T

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BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT S2406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 3 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION B = 50.000

LENGTH OF FOUNDATION L = 100.000

DEPTH OF EMBEDMENT , START VALUE ZSS = 25.000

DEPTH OF EMBEDMENT , FINAL VALUE ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE ZSSTEP = 5.000

SOIL TOTAL UNIT WEIGHT GAMS = 0.020

WATER UNIT WEIGHT GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE PVS = 500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE PVF = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE PVSTEP = 500.000

HORIZONTAL FORCE AT MUDLINE PH = 0.000

OVERTURNING MOMENT AT MUDLINE M = 0.000

EXCESS WATER PRESSURE OUTSIDE HEEL DELPH = 0.000

EXCESS WATER PRESSURE OUTSIDE TOE DELPT = 0.000

LOAD FACTORS TO BE USED LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL REDBS = 0.000

REDUCTION FACTOR SIDE, SOIL/SOIL REDSS = 0.500

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.050

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 2/ 2

PROJECT : EXAMPLE 3 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITION THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	25.0	6.87	51.00	29.2	0.0104	0.0108	0.0113
1.00	500.0	30.0	8.15	60.00	33.1	0.0103	0.0085	0.0090
1.00	1000.0	25.0	3.43	51.00	29.5	0.0208	0.0243	0.0254
1.00	1000.0	30.0	4.07	60.00	33.5	0.0207	0.0213	0.0218
1.00	1500.0	25.0	2.28	51.00	29.7	0.0312	0.0377	0.0396
1.00	1500.0	30.0	2.71	60.00	33.8	0.0311	0.0340	0.0357

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

C A P T

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 3 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	50.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	25.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSSTEP =	5.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.015
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	500.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.500

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EXAMPLE 3 ***CAPT***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

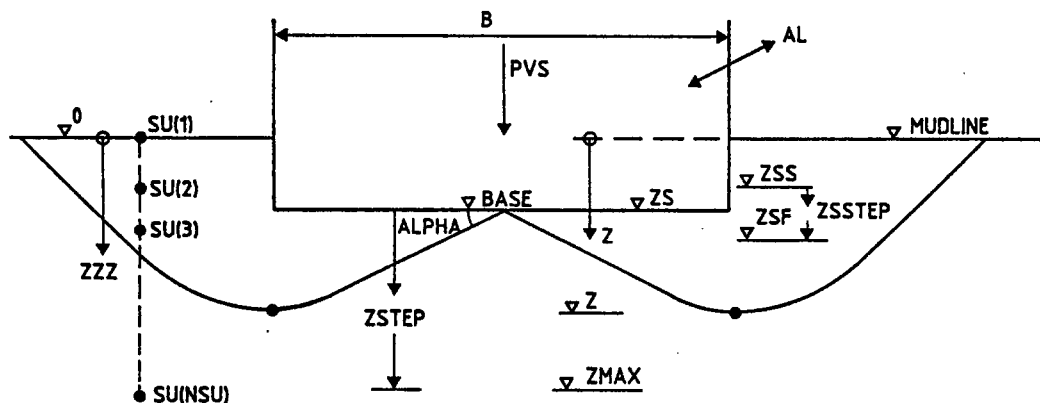
AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	25.0	7.13	53.00	32.0	0.0108	0.0121	0.0127
1.00	500.0	30.0	8.43	62.00	36.1	0.0106	0.0106	0.0112
1.00	1000.0	25.0	3.56	53.00	32.4	0.0216	0.0256	0.0269
1.00	1000.0	30.0	4.21	62.00	36.7	0.0213	0.0234	0.0245
1.00	1500.0	25.0	2.37	53.00	32.6	0.0324	0.0391	0.0410
1.00	1500.0	30.0	2.80	62.00	37.0	0.0320	0.0361	0.0379

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

INPUT
DATA FORM
FOR CAPT

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#1 MAY BE LEFT BLANK

C A P T

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 3 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION

B = 50.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 25.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSSTEP = 5.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.015

WATER UNIT WEIGHT

GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 500.000

HORIZONTAL FORCE AT MUDLINE

PH = 0.000

OVERTURNING MOMENT AT MUDLINE

M = 0.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.000

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = 0.000

LOAD FACTORS TO BE USED

LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.000

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.500

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW
MUDLINE

UNDRAINED
SHEAR STRENGTH

MINIMUM
SHEAR STRENGTH

0.000

0.001

0.001

100.000

0.225

0.050

PROJECT : EXAMPLE 5 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

SPECIFIED SURFACE CALCULATION

MINIMUM SAFETY FACTOR SF = 8.916
 FOUND FOR DEPTH Z = 45.000
 AND SLIP SURFACE ANGLE ALPHA = 15.857

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00
3	-RH	-0.1191E+02	-0.1750E+04	0.061	0.007	0.0000E+00
4	HA	0.6573E+03	-0.6088E+04	0.081	0.009	0.7125E+03
5	-PA	-0.2409E+02	-0.7712E+04	0.028	0.003	0.0000E+00
6	-HP	-0.6085E+03	-0.3856E+04	0.106	0.012	-0.5625E+03
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.003	0.0000E+00
8	-RS2	-0.7674E+01	-0.7757E+03	0.088	0.010	0.0000E+00
9	-RS3	-0.5049E+01	-0.1459E+04	0.031	0.003	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.002	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.002	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.002	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00

PROJECT : EXAMPLE 5 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

SPECIFIED SURFACE CALCULATION

MINIMUM SAFETY FACTOR SF = 7.937
 FOUND FOR DEPTH Z = 75.000
 AND SLIP SURFACE ANGLE ALPHA = 45.000

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00
3	-RH	-0.1338E+02	-0.1750E+04	0.061	0.008	0.0000E+00
4	HA	0.2650E+04	-0.1033E+05	0.125	0.016	0.2812E+04
5	-PA	-0.2706E+02	-0.7712E+04	0.028	0.004	0.0000E+00
6	-HP	-0.2543E+04	-0.1157E+05	0.124	0.016	-0.2363E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.004	0.0000E+00
8	-RS2	-0.5623E+02	-0.4097E+04	0.109	0.014	0.0000E+00
9	-RS3	-0.1062E+02	-0.2732E+04	0.031	0.004	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.002	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.003	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.003	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00

PROJECT : EXAMPLE 5 ***CAPT***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

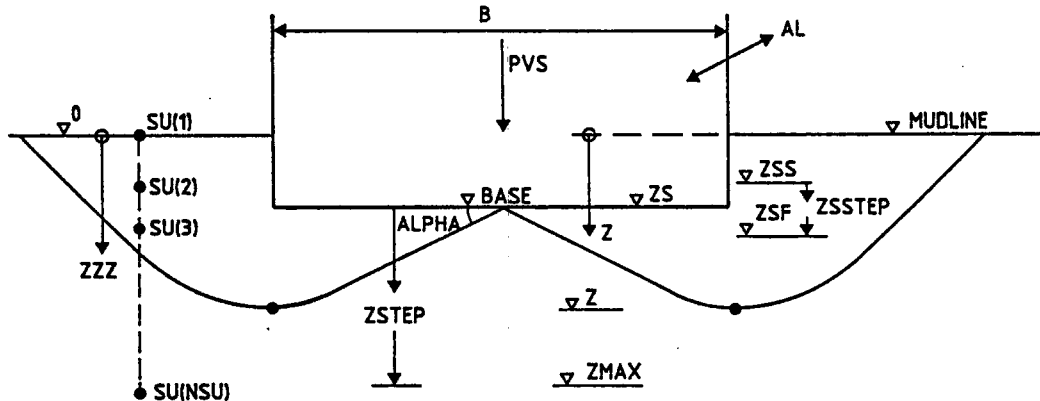
IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	7.47	62.00	35.7	0.0106	0.0085	0.0090

*)
 THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

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----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 6 ***CAPT***

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	50.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

1/24

PROJECT : EXAMPLE 6 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 8.088
 FOUND FOR DEPTH Z = 58.000
 AND SLIP SURFACE ANGLE ALPHA = 30.641

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00
3	-RH	-0.3397E+01	-0.1750E+04	0.016	0.002	0.0000E+00
4	HA	0.1420E+04	-0.7437E+04	0.100	0.012	0.1512E+04
5	-PA	-0.2656E+02	-0.7712E+04	0.028	0.003	0.0000E+00
6	-HP	-0.1330E+04	-0.7198E+04	0.110	0.014	-0.1232E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.004	0.0000E+00
8	-RS2	-0.4484E+02	-0.3727E+04	0.097	0.012	0.0000E+00
9	-RS3	-0.1534E+02	-0.4021E+04	0.031	0.004	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.002	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.003	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.003	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00

PROJECT : EXAMPLE 6 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	8.09	58.00	30.6	0.0088	0.0085	0.0090

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 6 ***CAPT***

THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	60.000
LENGTH OF FOUNDATION	L =	90.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EXAMPLE 6 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 9.258
 FOUND FOR DEPTH Z = 58.000
 AND SLIP SURFACE ANGLE ALPHA = 25.875

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00
3	-RH	-0.3205E+01	-0.1890E+04	0.016	0.002	0.0000E+00
4	HA	0.1264E+04	-0.7312E+04	0.099	0.011	0.1342E+04
5	-PA	-0.2088E+02	-0.6941E+04	0.028	0.003	0.0000E+00
6	-HP	-0.1186E+04	-0.6478E+04	0.110	0.012	-0.1109E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.003	0.0000E+00
8	-RS2	-0.4099E+02	-0.3955E+04	0.096	0.010	0.0000E+00
9	-RS3	-0.1340E+02	-0.4021E+04	0.031	0.003	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.003	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.002	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.002	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.002	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.004	0.0000E+00

PROJECT : EXAMPLE 6 ***CAPT***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	9.26	58.00	25.9	0.0076	0.0090	0.0094

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 6 ***CAPT***

THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	70.000
LENGTH OF FOUNDATION	L =	80.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EXAMPLE 6 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 10.237

FOUND FOR DEPTH Z = 60.000

AND SLIP SURFACE ANGLE ALPHA = 24.059

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.003	0.0000E+00
3	-RH	-0.3006E+01	-0.1960E+04	0.016	0.002	0.0000E+00
4	HA	0.1223E+04	-0.7267E+04	0.101	0.010	0.1294E+04
5	-PA	-0.1679E+02	-0.6170E+04	0.028	0.003	0.0000E+00
6	-HP	-0.1147E+04	-0.6170E+04	0.111	0.011	-0.1080E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.003	0.0000E+00
8	-RS2	-0.4354E+02	-0.4630E+04	0.096	0.009	0.0000E+00
9	-RS3	-0.1263E+02	-0.4191E+04	0.031	0.003	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.002	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.002	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.002	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.002	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.003	0.0000E+00

PROJECT : EXAMPLE 6 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	10.24	60.00	24.1	0.0071	0.0094	0.0098

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 6 ***CAPT***

THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	80.000
LENGTH OF FOUNDATION	L =	80.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EXAMPLE 6 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 12.122
 FOUND FOR DEPTH Z = 60.000
 AND SLIP SURFACE ANGLE ALPHA = 21.138

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.003	0.0000E+00
3	-RH	-0.2901E+01	-0.2240E+04	0.016	0.001	0.0000E+00
4	HA	0.1203E+04	-0.7893E+04	0.099	0.008	0.1267E+04
5	-PA	-0.1418E+02	-0.6170E+04	0.028	0.002	0.0000E+00
6	-HP	-0.1137E+04	-0.6170E+04	0.111	0.009	-0.1080E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.003	0.0000E+00
8	-RS2	-0.3846E+02	-0.4896E+04	0.095	0.008	0.0000E+00
9	-RS3	-0.1067E+02	-0.4191E+04	0.031	0.003	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.002	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.002	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.002	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.002	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.003	0.0000E+00

PROJECT : EXAMPLE 6 ***CAPT***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	12.12	60.00	21.1	0.0059	0.0086	0.0089

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 6 ***CAPT***

THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	90.000
LENGTH OF FOUNDATION	L =	80.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EXAMPLE 6 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 14.115
 FOUND FOR DEPTH Z = 62.000
 AND SLIP SURFACE ANGLE ALPHA = 20.154

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.002	0.0000E+00
3	-RH	-0.2803E+01	-0.2520E+04	0.016	0.001	0.0000E+00
4	HA	0.1293E+04	-0.8697E+04	0.101	0.007	0.1355E+04
5	-PA	-0.1217E+02	-0.6170E+04	0.028	0.002	0.0000E+00
6	-HP	-0.1230E+04	-0.6581E+04	0.113	0.008	-0.1178E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.002	0.0000E+00
8	-RS2	-0.3837E+02	-0.5653E+04	0.096	0.007	0.0000E+00
9	-RS3	-0.9531E+01	-0.4361E+04	0.031	0.002	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.002	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.001	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.001	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.001	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.002	0.0000E+00

PROJECT : EXAMPLE 6 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS A5ONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	14.12	62.00	20.2	0.0052	0.0078	0.0081

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 6 ***CAPT***

THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	80.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EXAMPLE 6 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 16.229

FOUND FOR DEPTH Z = 64.000

AND SLIP SURFACE ANGLE ALPHA = 19.348

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.002	0.0000E+00
3	-RH	-0.2709E+01	-0.2800E+04	0.016	0.001	0.0000E+00
4	HA	0.1388E+04	-0.9508E+04	0.103	0.006	0.1448E+04
5	-PA	-0.1059E+02	-0.6170E+04	0.028	0.002	0.0000E+00
6	-HP	-0.1328E+04	-0.6993E+04	0.114	0.007	-0.1278E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.002	0.0000E+00
8	-RS2	-0.3846E+02	-0.6465E+04	0.097	0.006	0.0000E+00
9	-RS3	-0.8612E+01	-0.4531E+04	0.031	0.002	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.001	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.001	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.001	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.001	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.002	0.0000E+00

PROJECT : EXAMPLE 6 ***CAPT***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	16.23	64.00	19.3	0.0047	0.0072	0.0074

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 6 ***CAPT***
 THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	110.000
LENGTH OF FOUNDATION	L =	80.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK
 PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EXAMPLE 6 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 18.448
 FOUND FOR DEPTH Z = 64.000
 AND SLIP SURFACE ANGLE ALPHA = 17.606

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.002	0.0000E+00
3	-RH	-0.2621E+01	-0.3080E+04	0.016	0.001	0.0000E+00
4	HA	0.1376E+04	-0.1019E+05	0.102	0.006	0.1433E+04
5	-PA	-0.9315E+01	-0.6170E+04	0.028	0.002	0.0000E+00
6	-HP	-0.1322E+04	-0.6993E+04	0.114	0.006	-0.1278E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.002	0.0000E+00
8	-RS2	-0.3521E+02	-0.6780E+04	0.096	0.005	0.0000E+00
9	-RS3	-0.7577E+01	-0.4531E+04	0.031	0.002	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.001	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.001	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.001	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.001	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.002	0.0000E+00

PROJECT : EXAMPLE 6 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	18.45	64.00	17.6	0.0041	0.0066	0.0067

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406-49 DATED 1 DEC 1985

PROJECT : EXAMPLE 6 ***CAPT***

THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-UNITS MN & M

WIDTH OF FOUNDATION	B =	120.000
LENGTH OF FOUNDATION	L =	80.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	0.000
OVERTURNING MOMENT AT MUDLINE	M =	0.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.000
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	0.000
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EXAMPLE 6 ***CAPT***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 20.786
 FOUND FOR DEPTH Z = 66.000
 AND SLIP SURFACE ANGLE ALPHA = 17.130

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
2	PA	0.0000E+00	0.0000E+00	0.035	0.002	0.0000E+00
3	-RH	-0.2538E+01	-0.3360E+04	0.016	0.001	0.0000E+00
4	HA	0.1477E+04	-0.1101E+05	0.104	0.005	0.1532E+04
5	-PA	-0.8267E+01	-0.6170E+04	0.028	0.001	0.0000E+00
6	-HP	-0.1424E+04	-0.7404E+04	0.116	0.006	-0.1382E+04
7	-RS1	0.0000E+00	0.0000E+00	0.031	0.001	0.0000E+00
8	-RS2	-0.3568E+02	-0.7669E+04	0.097	0.005	0.0000E+00
9	-RS3	-0.6976E+01	-0.4700E+04	0.031	0.001	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.001	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.001	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.001	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.001	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.002	0.0000E+00

PROJECT : EXAMPLE 6 ***CAPT***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	500.0	30.0	20.79	66.00	17.1	0.0037	0.0061	0.0062

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

	OPPDRAG : JOB TITLE:	OPPDRAGSNR. CONTRACT NO. 52406-48	SIDE PAGE II-0
	FORSØK / BEREGNING: SUBJECT:	SIGN. <i>RA</i>	DATO DATE 10/12-85
	<i>The Computer Program CAPT</i>	KONTR. CONTR.	DATO DATE

APPENDIX II

COMPARISON BETWEEN CAPT AND DAVIS & BOOKER

CONTENT	PAGE
COMPARISON F-factor	II-1
COMPARISON FAILURE ZONE	II-2

	OPPDAG : JOB TITLE:	OPPDRAGSNR. CONTRACT NO. 52406-48	SIDE PAGE II - 1
	FORSØK/BEREGNING: SUBJECT:	SIGN. <i>RAL</i>	DATO DATE 10/12-85
	CAPT	KONTR. CONTR.	DATO DATE

COMPARISON F - factor.

The following cases have been calculated by CAPT, and the corresponding "F_c" - factor has been back-calculated from:

$$\frac{Q}{B} = "F_c" \left[(2 + \pi) C_0 + \frac{\rho B}{4} \right]$$

B	ρ	C ₀	$\frac{\rho B}{C_0}$	Z	α	$\frac{Q}{B}$	"F _c "	ΔF
85	0.25	21	1.01	30	44.9	135	1.19	- 1%
85	0.5	21	2.02	26	37.7	156	1.31	- 2%
85	1.7	21	6.88	17	23.6	228	1.58	- 1%
85	4.0	21	16.2	12	16.4	335	1.73	0
40	3.61	21	6.88	8	23.6	229	1.585	- 1%
20	7.22	21	6.88	4	23.6	230	1.59	- 1%

The correlation between the two methods seems very good, for a wide range of $\frac{\rho B}{C_0}$. Differences of the order of 1% on the safe side as compared with Davis and Booker are found. Results seems not to be affected by changes in B.

Calculated values of "F_c" are plotted together with the Davis & Booker - curve on the next page.



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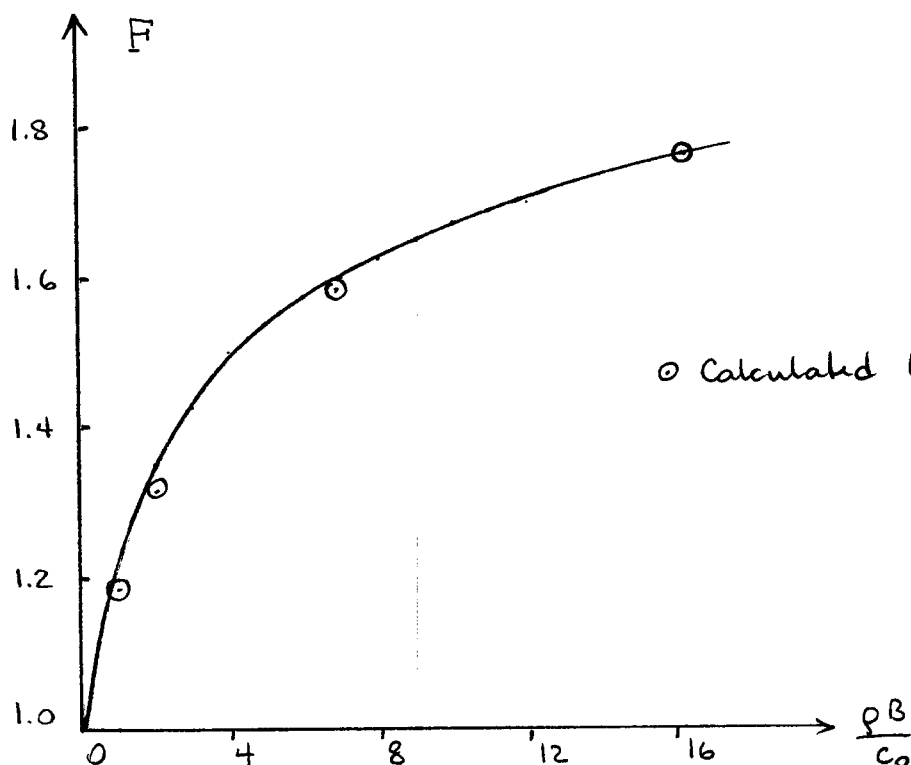
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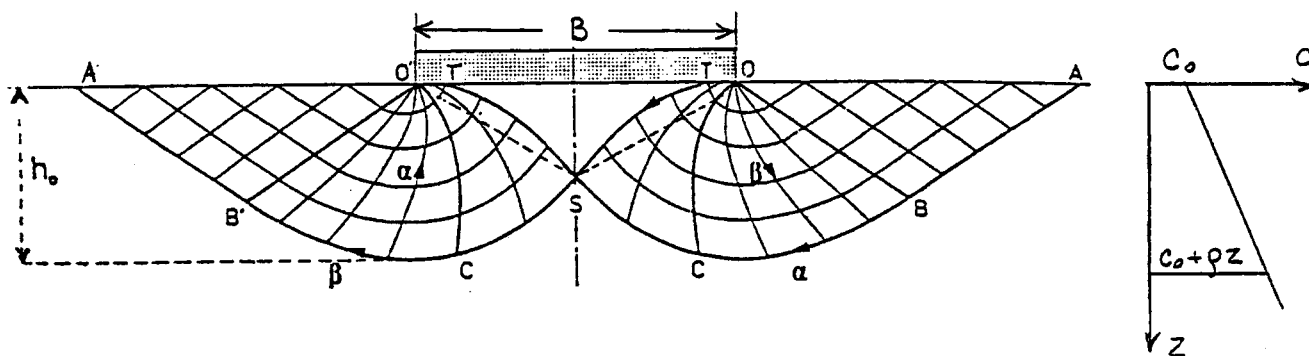
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COMPARISON of FAILURE ZONE

The shape of Davis & Booker's failure zone is also given by Matar & Salençon (1979).



The depth h_0 of the deepest point of the failure zone has been investigated.



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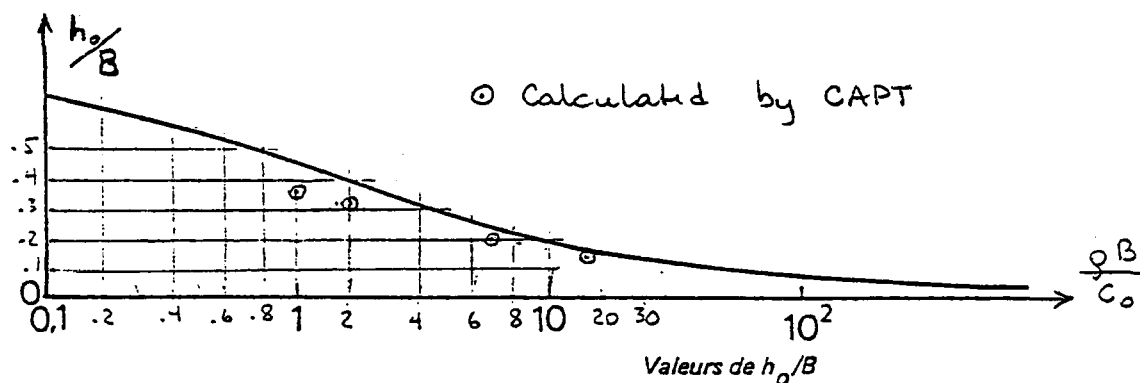
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The depth of the slip surface z in CAPT is compared to h_0 in 4 points as shown in the above figure.

$\frac{qB}{c_0}$	CAPS $\frac{z}{B}$	Matar & Salençon $\frac{h_0}{B}$
1	0.35	0.45
2	0.31	0.39
6.9	0.20	0.22
16	0.14	0.16

The slip surface in CAPT is shallower than the stress field of Davis & Booker. General behavior of CAPT slip surface looks satisfactory for surface load.

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APPENDIX III

CASE RECORD

CONTENT

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